

☐ ٤٥

6

* رقم الكمبيوتر

Enter your answer

7

* الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

Enter your answer

☒ Will be reviewed

8

OpenGL supports z-buffering.

☒ True

☐ False

☒ Will be reviewed

9

What is the difference between glVertex2f and glVertex3f?

☒ glVertex2f defines 2D vertices, while glVertex3f defines 3D vertices.

☐ glVertex2f uses integer coordinates, while glVertex3f uses floating-point coordinates.

☐ There is no difference between the two functions.

 **Will be reviewed**

10

Order the following shader processes:

1 `glCreateProgram`

2 `glLinkProgram`

3 `glAttachShader`

4 `glUseProgram`

1
3
2
4

 **Will be reviewed**

11

What is the purpose of the `glPolygonMode` function?

☐ To set the color of a polygon

☐ To set the texture of a polygon

☒ To set the mode in which polygons are rendered

☐ To set the depth test function

 **Will be reviewed**

12

What is the purpose of the `glutDisplayFunc()` callback?

- ☐ To handle window resizing events.
- ☐ To handle keyboard input events.
- ☐ To handle mouse input events.
- ☒ To define the rendering code that is called whenever the window needs to be redrawn.

 **Will be reviewed**

13

What is the primary purpose of culling in computer graphics?

- ☐ To improve the visual quality of rendered images
- ☐ To reduce the number of polygons that need to be rendered
- ☒ To reduce the frame rate of a graphics application
- ☐ To optimize the use of graphics hardware

 **Will be reviewed**

14

What are the parameters typically passed to the glutReshapeFunc callback function?

- ☒ The width and height of the window in pixels.
- ☐ The x and y coordinates of the mouse cursor.
- ☐ The current time in milliseconds.
- ☐ The state of the keyboard keys.

 Will be reviewed

15

When is the timer callback function registered with glutTimerFunc invoked?

- ☐ Immediately after glutTimerFunc is called.
- ☐ After a specified interval, in milliseconds.
- ☒ When the window is resized.
- ☐ When a key is pressed.

 Will be reviewed

16

How does glColor affect the appearance of primitives drawn after it is called?

- ☒ It sets the color for all subsequent primitives until a new glColor call is made.
- ☐ It only affects the color of the next primitive.
- ☐ It affects the color of all primitives in the scene.
- ☐ It only affects the color of the preceding primitive.

 Will be reviewed

17

When should glutMainLoop() typically be called in an OpenGL program?

- ☐ At the beginning of the program.

- ☒ After initializing OpenGL and setting up the display callback function.
- ☐ After drawing the scene.
- ☐ Before creating the window.

 **Will be reviewed**

18

Can you set the alpha component of a color using glColor?

- ☒ Yes.
- ☐ No.

 **Will be reviewed**

19

How many parameters does glViewport take?

- ☐ 2
- ☐ 3
- ☒ 4
- ☐ 5

 **Will be reviewed**

20

Backface culling refers to eliminating geometry with backfacing normals.

☒ True

☐ False

 **Will be reviewed**

21

OpenGL version 4.x does not support shaders.

☐ True

☒ False

 **Will be reviewed**

22

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently.

☒ True

☐ False

 **Will be reviewed**

23

Which matrix is used to project 3D objects onto a 2D screen?

☐ Modelview matrix

☒ Projection matrix

☐ Texture matrix

☐ Viewport matrix

 **Will be reviewed**

24

When is the glutReshapeFunc callback function invoked?

- ☐ When mouse is moved.
- ☒ When the window is resized.
- ☐ When a key is pressed.
- ☐ When the mouse is clicked.

 **Will be reviewed**

25

What is the relationship between glClear and glClearColor?

- ☐ glClearColor sets the color to clear the color buffer, while glClear actually clears the buffer.
- ☒ glClear sets the color to clear the color buffer, while glClearColor actually clears the buffer.
- ☐ Both functions are used to clear the color buffer, but with different parameters.
- ☐ There is no direct relationship between glClear and glClearColor.

 **Will be reviewed**

26

What is the primary function of glClipPlane?

- ☒ To define a clipping plane in 3D space.
- ☐ To clip a polygon against a rectangular region.
- ☐ To remove hidden surfaces from a 3D scene.
- ☐ To apply texture mapping to polygons.

 **Will be reviewed**

27

Which type of projection is commonly used for technical drawings and blueprints?

- ☒ Orthographic projection
- ☐ Perspective projection
- ☐ Metric projection
- ☐ Oblique projection

 **Will be reviewed**

28

Texture mapping **with** mipmapping consumes more memory than texture mapping **without** mipmapping.

- ☐ True
- ☒ False

 **Will be reviewed**

29

The **cross** product of two vectors that are orthogonal is the **zero** vector.

☒ True

☐ False

 **Will be reviewed**

30

What is the purpose of the `glFlush()` function in OpenGL?

☐ To clear the color buffer

☒ To force the execution of OpenGL commands

☐ To set the current color

☐ To define the viewing volume

 **Will be reviewed**

31

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(180.0, 0, 0, 1.0);  
glScalef(-1.0, 1.0, 1.0);
```

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☐ A

☐ B

☒ C

☐ D

 **Will be reviewed**

32

Which type of projection is commonly used in video games and 3D simulations?

☒ Orthographic projection

☐ Perspective projection

☐ Metric projection

☐ Oblique projection

 **Will be reviewed**

33

In modern OpenGL, the user may supply shaders which will execute on the GPU.

☐ True

☒ False

 **Will be reviewed**

34

Select the output of executing the following OpenGL code segment:

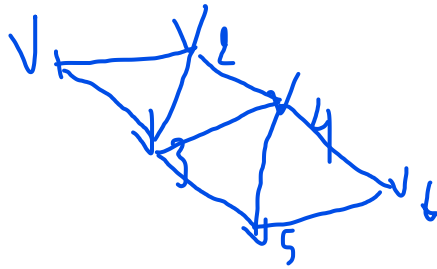
```
glBegin(GL_TRIANGLE_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

☐ A

☐ B

☒ C

☐ D



 **Will be reviewed**

35

What is the difference between orthographic and perspective projection?

- ☒ Orthographic projection preserves parallel lines, while perspective projection does not.
- ☐ Perspective projection preserves parallel lines, while orthographic projection does not.
- ☐ Orthographic projection uses a single vanishing point, while perspective projection uses multiple vanishing points.
- ☐ Perspective projection uses a single vanishing point, while orthographic projection uses multiple vanishing points.

 **Will be reviewed**

36

What happens inside the `glutMainLoop()` function?

- ☐ It continuously checks for events like keyboard input, mouse input, and window resizing.
- ☐ It calls the registered display callback function to redraw the scene.
- ☐ It handles window management tasks like creating, destroying, and resizing windows.

☒ All of the above.

 Will be reviewed

37

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-2.0, 1.0, -1.0);  
glRotatef(180.0, 0, 0, 1.0);
```

☐ A

☐ B

☐ C

☒ D

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

 Will be reviewed

38

How many parameters does glRectf take?

☐ 2

☒ 4

☐ 6

☐ 8

☑ Will be reviewed

39

In the following code segment using OpenGL Shading Language (GLSL), what is the value d?

```
vec4 TestVec = vec4(1.0, 2.0, 3.0, 4.0);  
vec2 d = TestVec.xz;
```

- ☐ d= NULL
- ☐ d=(3.0, 1.0)
- ☐ d=(1.0, 2.0, 3.0, 4.0)
- ☐ d=(1.0, 3.0)
- ☐ d=(4.0, 2.0)
- ☒ syntax error

☑ Will be reviewed

40

Specular reflections are view independent.

- ☐ True
- ☒ False

☑ Will be reviewed

41

How is a clipping plane defined in OpenGL?

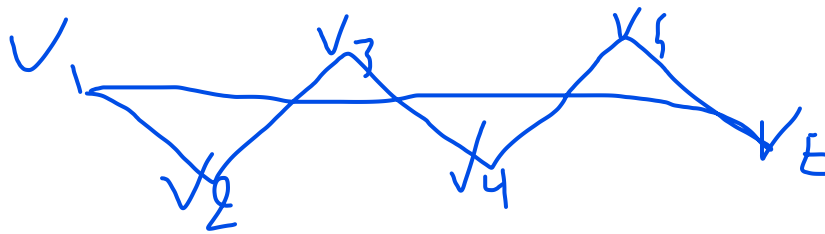
- ☐ By specifying four points that lie on the plane.
- ☒ By specifying the equation of the plane in the form $Ax + By + Cz + D = 0$.
- ☐ By specifying the normal vector to the plane and a point on the plane.
- ☐ By specifying the distance from the origin to the plane.

 **Will be reviewed**

42

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);
glVertex2fv(v1); glVertex2fv(v2);
glVertex2fv(v3); glVertex2fv(v4);
glVertex2fv(v5); glVertex2fv(v6);
glEnd();
```

- ☐ A 
- ☐ B
- ☐ C
- ☒ D
- ☐ E
- ☐ F

 **Will be reviewed**

43

What do the parameters of glRectf represent?

- ☐ The center point and radius of the rectangle.
- ☒ The top-left and bottom-right corners of the rectangle.
- ☐ The width and height of the rectangle.
- ☐ The color of the rectangle.

📝 **Will be reviewed**

44

Can glViewport be used to create multiple viewports within a single window?

- ☒ Yes.
- ☐ No.

📝 **Will be reviewed**

45

When should glClear be called in a typical OpenGL rendering pipeline?

- ☒ Before drawing any objects.
- ☐ After drawing all objects.
- ☐ Between drawing different objects.
- ☐ It doesn't matter when glClear is called.

📝 **Will be reviewed**

46

In OpenGL Shading Language, **attribute** parameters are passed from vertex shader to fragment shader.

☒ True

varying

☐ False

 **Will be reviewed**

47

What is the role of the glBegin and glEnd functions in relation to glVertex?

☒ They define the type of primitive to be drawn (e.g., points, lines, triangles).

☐ They specify the color of the vertices.

☐ They set the projection and modelview matrices.

☐ They control the depth testing and blending modes.

 **Will be reviewed**

48

What is the primary function of glRectf in OpenGL?

☒ Draws a filled rectangle.

☐ Draws a line.

☐ Draws a point.

☐ Clears the screen.

 Will be reviewed

49

What is the primary function of glVertex in OpenGL?

- ☒ To define a vertex in 3D space.
- ☐ To draw a line between two points.
- ☐ To set the color of vertices.
- ☐ To set the color of subsequent vertices.

 Will be reviewed

50

The **positional light source** is appropriate for a distant light source, e.g., **the sun**.

- ☐ True
- ☒ False

 Will be reviewed

51

OpenGL Shading Language (GLSL) supports pointers.

- ☒ True
- ☐ **False**

 Will be reviewed

52

The parameters of the glViewport function typically specify:

- ☐ a) The width and height of the viewport
- ☐ b) The x and y coordinates of the viewport's lower-left corner
- ☒ c) Both (a) and (b)
- ☐ d) None of the above

 Will be reviewed

53

What is the primary function of glutTimerFunc in OpenGL?

- ☐ To initialize the OpenGL environment.
- ☐ To create a window on the screen.
- ☐ To enter the main event processing loop.
- ☒ To register a timer callback function.

 Will be reviewed

54

Which of the following is NOT a valid mode for glPolygonMode?

- ☐ GL_POINT

- ☐ GL_LINE
- ☐ GL_FILL
- ☒ GL_WIREFRAME

📝 Will be reviewed

55

How many parameters does `glutTimerFunc` take?

- ☐ 1
- ☒ 2
- ☐ 3
- ☐ 4

📝 Will be reviewed

56

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

- ☐ A
- ☐ B
- ☒ C



☐ D

☐ E

☐ F

 **Will be reviewed**

57

What is the difference between GL_LINES and GL_LINE_STRIP?

- ☒ GL_LINES draws individual lines, while GL_LINE_STRIP connects vertices with lines.
- ☐ GL_LINES connects vertices with lines, while GL_LINE_STRIP draws individual lines.
- ☐ GL_LINES draws a closed polygon, while GL_LINE_STRIP draws an open polygon.
- ☐ There is no difference between the two.



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